

Question ID: 9f934297

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right rectangular prism has a length of **28 centimeters (cm)**, a width of **15 cm**, and a height of **16 cm**. What is the surface area, **in cm²**, of the right rectangular prism?

Correct Answer: 2216

Rationale

The correct answer is **2,216**. The surface area of a prism is the sum of the areas of all its faces. A right rectangular prism consists of six rectangular faces, where opposite faces are congruent. It's given that this prism has a length of **28 cm**, a width of **15 cm**, and a height of **16 cm**. Thus, for this prism, there are two faces with area **(28)(15) cm²**, two faces with area **(28)(16) cm²**, and two faces with area **(15)(16) cm²**. Therefore, the surface area, **in cm²**, of the right rectangular prism is **2(28)(15) + 2(28)(16) + 2(15)(16)**, or **2,216**.